Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Quantum pure states are not enough

Files.

- File: `briefs/PureStatesNotEnough/PureStatesNotEnough.lean`
- Library: `PureStatesNotEnough`
- Imports: one new `ForMathlib` file:
  `ForMathlib/Analysis/Convex/Combination.lean`: a map that preserves the convex combinations of two points of a convex set preserves all finite convex combinations (`Convex.map_sum_of_map_add`), and a convex set contains them (`Convex.sum_mem_of_semifield`), with scalars in any ordered semifield such as $\mathbb{R}_{\geq 0}$.
  Existing files: `ForMathlib/Analysis/InnerProductSpace/l2Space.lean` (`Orthonormal.countable`), `ForPhyslib/QuantumMechanics/DensityOperator.lean` (`IsDensityOperator`, `posTrace`), `ForPhyslib/QuantumMechanics/Wigner/Density.lean` (`IsDensity`, `proj`) and `ForPhyslib/ProbabilisticTheory/HilbertSpace/TraceClass/CanonicalExpansion.lean` (`IsTraceClass.exists_hasSum_rankOne`).

Setting. The ensembles of CST over a measurable space X with Liouville measure μ are the probability measures $\nu \ll \mu$ (`classicalEnsembles` μ). Their mixtures are $a\nu_1 + b\nu_2$ with $a, b \in \mathbb{R}_{\geq 0}$, $a + b = 1$. `HasManyDisjointSets` μ says that for every N there are N disjoint measurable sets of positive finite measure. The density operators of QE-DENS are `IsDensity` (positive, trace one) on a finite-dimensional space and `IsDensityOperator` (positive, trace class, trace one) on a separable complex Hilbert space $\mathcal{H}$ of any dimension. The pure states (`pureStates`) are the projections $|\psi\rangle\langle\psi|$ with $\|\psi\| = 1$, a copy of $P(\mathcal{H})$ inside the bounded operators, with the Borel σ-algebra of the operator norm. An identification of two ensemble spaces is modelled as a map that preserves mixtures of two ensembles.

Results.

Brief	Lean
CST, the ensemble space	`classicalEnsembles`; it is convex: `Convex.classicalEnsembles`
(used for Theorem 3) a Darboux chart has N disjoint sets of positive finite Liouville measure for every N	`hasManyDisjointSets_restrict`: any nonempty open set in a nontrivial finite-dimensional real normed space, for any measure that is positive on open sets and finite on compact sets
(used for Theorem 3) the CST ensemble space is infinite-dimensional	`not_injOn_of_map_add`: no mixture-preserving map from it into a finite-dimensional real vector space is injective
Theorem 3, CST and QE-DENS are inconsistent	`cst_qeDens_inconsistent`, for $\dim \mathcal{H} < \infty$
Theorem 1, QP-RAY and CST are consistent iff $1 \leq \dim V < \infty$	not formalized; `subsingleton_projectivization`: for $\dim V = 1$, $P(V)$ is a single point
Remark after Example 2, the Born rule is not $\delta_{\psi\phi}$	`born_ne_delta`: e_0 and $(e_0 + e_1)/\sqrt{2}$ are distinct rays with Born probability $\frac{1}{2}$
Theorem 5, the map $r : M^1(P(\mathcal{H})) \to D(\mathcal{H})$	`barycenter` $r(\mu) = \int \rho \, d\mu(\rho)$; `isDensityOperator_barycenter`, `posTrace_barycenter`; `toDensity`
Theorem 5, r is affine	`barycenter_add`
Theorem 5, $D(\mathcal{H}) \equiv M^1(P(\mathcal{H}))/\sim$	`toDensity_surjective`, `densityEquiv`
Remark after Theorem 5, a class records the different preparations of one ensemble	`exists_barycenter_eq_and_ne`: $\frac{1}{2}\delta_{e_0} + \frac{1}{2}\delta_{e_1}$ and $\frac{1}{2}\delta_{e_+} + \frac{1}{2}\delta_{e_-}$ have the same barycenter but give e_0 different probabilities

Differences from the brief.

- Theorem 3 compares the whole ensemble spaces, not those supported on a two-dimensional subspace A (see "Problems found"). It assumes $\dim \mathcal{H} < \infty$, which the brief derives from Theorem 1. Only the finite dimension of the real vector space of operators is used, not positivity or trace. "The spaces correspond" becomes a map from the CST ensembles to the density operators that is injective and preserves mixtures of two ensembles. The proof normalizes μ on $\dim + 2$ disjoint sets, takes an affine dependence among the images, and separates its positive and negative parts into two different mixtures with the same image.
- The Liouville measure enters only through `HasManyDisjointSets`, which holds on every Darboux chart. The symplectic structure itself is not used.

- Theorem 5 is proved for every separable complex Hilbert space, not only in finite dimension. $r(\mu)$ is the Bochner integral of the pure states in the operator norm. It is positive because the positive operators form a closed convex cone, and it has trace one by Parseval and monotone convergence over a countable Hilbert basis. r is onto by the canonical expansion $\rho = \sum_i \lambda_i |e_i\rangle\langle e_i|$, represented by $\sum_i \lambda_i \delta_{e_i}$. No Choquet theory is used. "$\equiv$" is the bijection between the quotient by $r(\mu_1) = r(\mu_2)$ and $D(\mathcal{H})$.
- Not formalized:
  - Theorem 1;
  - Example 2: $P(\mathbb{C}^2)$ is a sphere with Darboux coordinates z, θ_{xy}; Hamiltonian flows; quantum evolution as rotations;
  - Insight 4, which is informal;
  - that the pure states are the extreme points of $D(\mathcal{H})$, which the proof of Theorem 5 does not need.

Problems found in the brief. Theorems 3 and 5 hold, but their proofs have gaps. Theorem 1 is not checked.

- Theorem 1 states $1 \leq \dim V$, while its proof uses $2 \leq \dim X$. For $\dim V = 1$ the state space $P(V)$ is a single point (`subsingleton_projectivization`), a manifold of dimension 0. So either 0-dimensional symplectic manifolds count, and the proof's bound should be $0 \leq \dim X$, or they do not, and the theorem should say $2 \leq \dim V < \infty$. "Since X is a symplectic manifold, $\dim X < \infty$" holds only for finite-dimensional symplectic manifolds, and the projective space of an infinite-dimensional Hilbert space is an infinite-dimensional one. The finiteness comes from CST itself: the Liouville measure ω^n needs a finite n.
- In the proof of Theorem 3, for $\dim \mathcal{H} \geq 3$ the sphere $P(A)$ has dimension $2 < 2(\dim \mathcal{H} - 1)$ and therefore Liouville measure zero. No CST ensemble has support in $P(A)$, so the classical side is empty, not infinite-dimensional. The argument works as written only for $\dim \mathcal{H} = 2$. Comparing the whole spaces repairs it.
- The proof of Theorem 5 cites Choquet theory, which concerns compact convex sets, but $D(\mathcal{H})$ is not compact when $\dim \mathcal{H} = \infty$. The proof also never shows that every density operator is represented, which $D(\mathcal{H}) \equiv M^1/\sim$ needs. Both gaps are closed by the canonical expansion. "Every continuous linear functional $f : D(\mathcal{H}) \to \mathbb{R}$" should read continuous affine function, since $D(\mathcal{H})$ is not a vector space.
- Example 2 (not checked in Lean): area-preserving diffeomorphisms of the sphere include the orientation-reversing ones, such as $z \mapsto -z$, which are not Hamiltonian. So Hamiltonian evolution preserves the area form, not only the area.
- Typos: "non necessarily" should be "not necessarily"; "r defined about" should be "r defined above".